

Meihao Fan

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Scalable Data Synthesis for Agent Training

Summer 2027 U.S. Research Internship | Embodied AI Data, World Models & Multimodal Agents

Research Summary

Ph.D. student at Renmin University of China researching **scalable data synthesis for agent training**. I develop methods for **task generation, interaction-trajectory synthesis, and execution-based data verification**, and use the resulting data for curriculum learning and multi-turn reinforcement learning. My work connects data construction to measurable improvements in agent capability and generalization. I am extending this research to **embodied data curation, action-conditioned world models, and evaluation-driven data evolution**.

Education

Renmin University of China

Ph.D. in Computer Application Technology; Advisors: Prof. Ju Fan and Prof. Xiaoyong Du
National Scholarship (2025): the college's only second-year Ph.D. recipient.

Sept 2023 – Expected June 2028

Beijing, China

Chongqing Jiaotong University

B.S. in Computer Science and Technology; GPA: 4.31/5.00

National Scholarship (2023): the college's only recipient; Mingde Scholarship Nomination Award (university top 20).

Sept 2019 – June 2023

Chongqing, China

Research Experience

EvoPhys

Research Intern, World Models

Sept 2026 – Present

Beijing, China

- Lead research on **world model training**, spanning interactive-memory evaluation, scalable multimodal data annotation, and evaluation-driven recursive self-improvement (RSI).

ByteDance

Research Intern, LLM Agent Systems & Reinforcement Learning

March 2025 – June 2026

Beijing, China

- Led DeepPrep, covering synthetic training tasks, an executable data-preparation environment, and progressive SFT/RL; led research on agent harness self-evolution from execution trajectories.

Renmin University of China

Research Assistant

Sept 2023 – March 2025

Beijing, China

- Led AutoPrep and BATCHER research on task-aware data preparation, multi-agent orchestration, demonstration selection, and cost-efficient LLM inference.

Selected Research Projects

DeepAnalyze: Scalable Training-Trajectory Synthesis

ICML 2026

Project Lead | Synthetic data, quality filtering, curriculum learning, agentic RL

- Developed **data-grounded trajectory synthesis** for DeepAnalyze-8B, combining teacher-distilled reasoning and environment-generated interactions in the **DataScience-Instruct-500K** training corpus.
- Built a **questioner-solver-inspector** pipeline: generate tasks and acceptance criteria from real data, execute multi-turn solutions, and filter trajectories using interaction checks and resulting environment changes.
- Implemented a curriculum from single-skill SFT to multi-skill cold-start training and **GRPO**, using approximately 470K, 20K, and 15K samples, respectively.
- Evaluated on **12 benchmarks**; achieved **38.88% on DABStep** and **61.7% on DS-1000**, versus 15.34% and 54.8% for the single-ability training variant.

DeepPrep: Verifiable Task Synthesis for Agent Training

VLDB 2026

Project Lead; First Author | Executable supervision, reversible corruption, multi-turn RL

- Designed a synthesis method that converts SQL workloads into **source tables, target specifications, and executable pipelines**; adds **reversible noise** and verifies recovery to preserve task solvability.
- Constructed **6,788 training tasks**, with pipelines spanning **31 operator types** and up to **28 steps**; trained agents through operator-level SFT, verified reasoning trajectories, and GRPO with hybrid rewards.
- Synthetic training data improved **out-of-domain accuracy by 37.0 percentage points** over Parrot training on Synth-Bird (Qwen3-8B, data ablation); tree-based execution supports backtracking and pipeline repair.

- Proposed an **interactive-memory benchmark for action-conditioned world models**, formalizing memory through **Encoding, Maintain, Update, and Read** and evaluating reactive response, memory persistence, and memory plasticity across temporal-retention and interference settings.
- Built an automated **Isaac Sim** evaluation pipeline with programmatically generated cases and ground-truth state supervision; reproduced and evaluated **nine representative world models**, including Ctrl-World, WorldMem, iVideoGPT, HyDRA, and Oasis, under a unified evaluation protocol.
- Built a **fully automated multimodal annotation system** for world-model training, covering atomic action segmentation, semantic and object-interaction labels, object-state changes, camera pose, and 3D hand motion; supports **100+ videos concurrently** at approximately **\$40 API cost per raw video hour**, with annotation quality validated through manual audits.
- Developing a **Recursive Self-Improvement** pipeline linking data annotation, training-recipe construction, distributed training, evaluation, and recipe refinement; uses model weaknesses to drive hard-example mining, curriculum construction, active data selection, and data-mixture optimization.

Additional Research Projects

CoDA-Bench & TACO: Scalable, Verifiable Task Construction

ICML / VLDB 2026

Co-author | Task synthesis, execution verification, difficulty refinement

- CoDA-Bench**: Co-authored a benchmark of **1,009 tasks across 31 data communities**, built from executable notebook solutions through task reconstruction, adversarial refinement, and human verification.
- TACO**: Co-authored a benchmark combining **1,500 expert-annotated** and **13,000 synthetic** NL-SQL pairs, using query-structure sampling, schema-constrained generation, and quality checks.

EvoCost & SkillAdam: Agent Self-Evolution

ICLR 2027 submissions

EvoCost: Lead, First Author | SkillAdam: Co-author

- EvoCost**: Developed harness evolution that mines avoidable interactions from trajectory dependency graphs and validates updates to instructions, skills, tools, and executor logic with a fixed backbone model.
- SkillAdam**: Co-authored skill optimization with persistent issue memory and adaptive edit budgets; reduced optimization tokens by **67.3%** versus SkillOpt on DeepPlanning, with test accuracy improving from 21.7% to 28.3%.

AutoPrep & BATCHER: Task-Aware Data Preparation

VLDB 2025 / ICDE 2024

Project Lead; First Author | Data transformation, demonstration selection, inference efficiency

- AutoPrep**: Developed Chain-of-Clauses planning and tool-augmented execution; improved NL2SQL accuracy by **12.22 / 13.23 points** on WikiTQ / TabFact, averaged across evaluated backbones.
- BATCHER**: Developed coverage-based demonstration selection and token-aware batch allocation; batch prompting reduced API cost by **4–7×** versus standard prompting across eight ER datasets.

Publications

- [1] **Meihao Fan**, Ju Fan, Yuxin Zhang, Shaolei Zhang, Xiaoyong Du, Jie Song, Peng Li, Fuxin Jiang, Tieying Zhang, Jianjun Chen. *DeepPrep: An LLM-Powered Agentic System for Autonomous Data Preparation*. **VLDB 2026**. Paper
- [2] **Meihao Fan**, Ju Fan, Nan Tang, Lei Cao, Guoliang Li, Xiaoyong Du. *AutoPrep: Natural Language Question-Aware Data Preparation with a Multi-Agent Framework*. **VLDB 2025**. Paper
- [3] **Meihao Fan**, Xiaoyue Han, Ju Fan, Chengliang Chai, Nan Tang, Guoliang Li, Xiaoyong Du. *Cost-Effective In-Context Learning for Entity Resolution: A Design Space Exploration*. **ICDE 2024**. Paper
- [4] Shaolei Zhang, Ju Fan, **Meihao Fan**, Guoliang Li, Xiaoyong Du. *DeepAnalyze: Agentic Large Language Models for Autonomous Data Science*. **ICML 2026**. Paper | Code
- [5] Yuxin Zhang, **Meihao Fan**, Ju Fan, Mingyang Yi, Yuyu Luo, Jian Tan, Guoliang Li. *Reward-SQL: Boosting Text-to-SQL via Stepwise Execution-Aware Reasoning and Process-Supervised Rewards*. **SIGMOD 2026**. Paper
- [6] Yuxin Zhang, Ju Fan, **Meihao Fan**, Shaolei Zhang, Xiaoyong Du. *CoDA-Bench: Can Code Agents Handle Data-Intensive Tasks?* **ICML 2026**. Paper
- [7] Chao Deng, Ju Fan, Yuyu Luo, Qinliang Xue, **Meihao Fan**, Yuxin Zhang, Min Zhang, Xiaofeng Jia, Jing Zhang, Xiaoyong Du. *TACO: A Benchmark for Open-Domain Text-to-SQL with Ambiguous and Cross-Database Queries*. **VLDB 2026**. Paper

Manuscripts

- [U1] **Meihao Fan**, Shaolei Zhang, Ju Fan, Peng Li, Jie Song, Jianjun Chen. *EvoCost: Harness Self-Evolution for Cost-Efficient LLM Agents*. **Under Review at ICLR 2027**.
- [U2] Gaoyuan Li, **Meihao Fan**, Yizhe Liu, Shaolei Zhang, Ju Fan, Siyi Wang, Jiaheng Hou, Xudong Weng, Honghan Tian, Zang Li. *SkillAdam: Stable and Efficient Skill Evolution for Agents*. **Under Review at ICLR 2027**. Preprint | Code
- [U3] Yuxin Zhang, Ju Fan, **Meihao Fan**, Shaolei Zhang, *Multi-Agent World: Scaling Multi-Agent Orchestration Training*

Technical Skills & Research Interests

Data: Task and trajectory synthesis, execution-based verification, quality filtering, demonstration selection.

Training: Supervised fine-tuning, GRPO, multi-turn RL, curriculum learning, reward modeling.

Tools: Python, PyTorch, Transformers, vLLM. **Languages:** Chinese (native), English (professional).

Research interests: Synthetic data for agents; embodied data; world models; multimodal agents; self-improving systems.